

SECURITY ENGINEERING — HANDS-ON LAB

CrowdStrike Falcon API Detection & Triage

Sensor → API Client → Real Detection → Automated Triage

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github.com/davidrufinobotte/security-engineering-labs

HOW IT WORKS

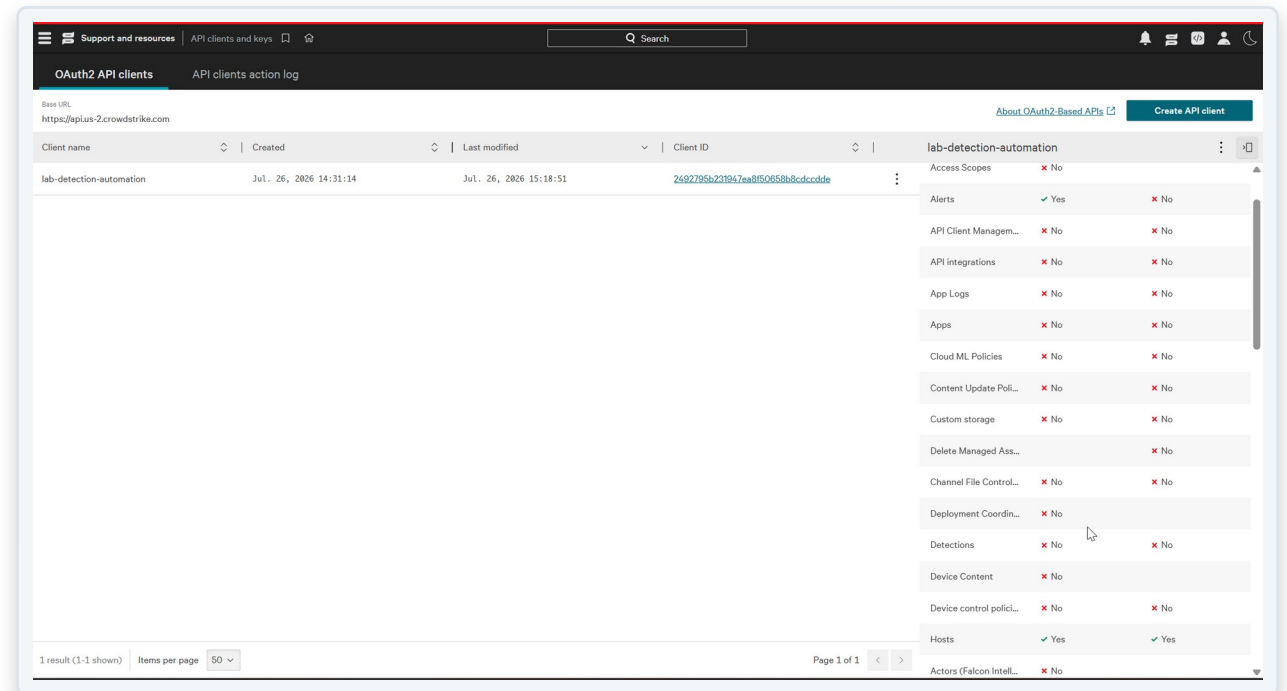
From endpoint sensor to automated triage



A safe, authorized simulated attack triggers real Falcon detections. A Python script reads them via API, then writes back a status update and comment — closing the loop from detection to response.

A Least-Privilege API Client

- Created a dedicated OAuth2 API client (lab-detection-automation)
- Scoped to exactly what the project needs: Alerts and Hosts — nothing else
- Same least-privilege principle applied earlier in the Entra ID + Sentinel lab



Sensor Deployed & Reporting

- Falcon sensor installed on a dedicated test laptop
- Confirmed online, managed, healthy, and up to date within minutes
- This is the data source everything downstream depends on

Falcon Asset Details | DRB-PF312QGZ

DRB-PF312QGZ
Windows | LENOVO | 20U9007RAU
Asset ID 137e3b40aa59408e97be921d67fe2fda
IP 10.2.0.2, 149.19.31.128

Asset management **Response actions**

Online
Connectivity status

Managed
Sensor coverage

Not registered
Cloud account

Not exposed
Internet exposure

Findings powered by Falcon Surface

Healthy
Sensor health

Up to date
Sensor version

Supported
OS support status

Up to date
Asset criticality

General

Account name	--
Account ID	--
Resource ID	--
Cloud provider	--
Cloud account	--
Region	--
Sensor IP address	149.19.31.128
MAC address	58-6c-25-ed-f9-0e
Status	--
Updated	Jul. 26, 2026 15:11:29
Instance State	--
Data providers	Falcon sensor, Active Directory
Discovering by	--

Hit a Decommissioned API

- First script used FalconPy's classic Detects service class — the industry-standard starting point
- CrowdStrike retired this API entirely; the request failed with a clear 404

```
PS C:\Users\DavidBotte\OneDrive - MSFT\Falcon> & C:\Users\DavidBotte\AppData\Local\Microsoft\WindowsApps\python3.13.exe "c:/Users/DavidBotte/OneDrive - MSFT/Falcon/falcon_list_detections.py"  
Failed to query detections: [{'code': 404, 'message': 'API endpoint has been decommissioned as per TA: https://supportportal.crowdstrike.com/s/article/Tech-Alert-Planned-Decommission-Announcement-of-the-DetectionSummaryEvent-and-detects-API'}]
```

Migrated API, Wrong Scope

- Rewrote the script against the current Alerts API — but the client's old scope name didn't carry over
- 403 access denied — fixed by granting the correct Alerts: Read scope

```
PS C:\Users\DavidBotte\OneDrive - MSFT\Falcon> & C:\Users\DavidBotte\AppData\Local\Microsoft\WindowsApps\python3.13.exe "c:/Users/DavidBotte/OneDrive - MSFT/Falcon/falcon_list_alerts.py"  
Failed to retrieve alerts: [{'code': 403, 'message': 'access denied, scope not permitted'}]
```

Confirmed Working — Before Any Data

- With auth and scope fixed, the script ran cleanly and correctly reported zero alerts
- Proves the pipeline works end to end before relying on it to catch something real

```
PS C:\Users\DavidBotte\OneDrive - MSFT\Falcon> & C:\Users\DavidBotte\AppData\Local\Microsoft\WindowsApps\python3.13.exe "c:/Users/DavidBotte/OneDrive - MSFT/Falcon/falcon_list_alerts.py"
No alerts found yet. This is expected on a fresh trial with no test activity.
PS C:\Users\DavidBotte\OneDrive - MSFT\Falcon> & C:\Users\DavidBotte\AppData\Local\Microsoft\WindowsApps\python3.13.exe "c:/Users/DavidBotte/OneDrive - MSFT/Falcon/falcon_list_alerts.py"
No alerts found yet. This is expected on a fresh trial with no test activity.
```

GENERATING A REAL SIGNAL

Why EICAR didn't work — and what did

EICAR: no detection.

Falcon is a behavior-based EDR — a static test file sitting on disk, never executed, simply doesn't trigger it. Expected, not a misconfiguration.

Atomic Red Team: real detection.

Ran MITRE technique T1053.005 (Scheduled Task persistence) — a safe, reversible simulation of real attacker behavior, with automatic cleanup after.

10 Alerts, Multiple Detection Layers

- One safe technique triggered detections across Execution, Machine Learning, and Malware categories — confirming several of Falcon's engines are active, not just one signature check

```
PS C:\Users\DavidBotte\OneDrive - MSFT\Falcon> & C:\Users\DavidBotte\AppData\Local\Microsoft\WindowsApps\python3.13.exe "c:/Users/DavidBotte/OneDrive - MSFT/Falcon/falcon_list_alerts.py"

Found 10 alert(s):

Hostname          Severity  Status    Created                                Tactic
-----
DRB-PF312QGZ      Medium    new       2026-07-26T07:14:37.215087618Z Execution
DRB-PF312QGZ      Medium    new       2026-07-26T07:14:37.215720815Z Execution
DRB-PF312QGZ      High      new       2026-07-26T07:14:36.034115457Z Machine Learning
DRB-PF312QGZ      Low       new       2026-07-26T07:14:36.031553007Z Malware
DRB-PF312QGZ      Medium    new       2026-07-26T07:14:36.022120312Z Machine Learning
DRB-PF312QGZ      High      new       2026-07-26T07:14:33.529206329Z Machine Learning
DRB-PF312QGZ      High      new       2026-07-26T07:14:33.527696006Z Machine Learning
DRB-PF312QGZ      Low       new       2026-07-26T07:14:32.657121195Z Malware
DRB-PF312QGZ      High      new       2026-07-26T07:14:32.656817272Z Machine Learning
DRB-PF312QGZ      Low       new       2026-07-26T07:14:32.649714135Z Malware

PS C:\Users\DavidBotte\OneDrive - MSFT\Falcon>
```

Confirmed in the Console Too

- Cross-checked the API results against the Falcon console directly on the host
- 17 total detections found — the API view and the console view agree
- Good practice: never trust a script's output alone when a source of truth is one click away

The screenshot displays the Falcon console interface for asset DRB-PF312QGZ. The left sidebar shows asset management options like 'Online', 'Not registered', and 'Not exposed'. The main panel is divided into 'Overview', 'Threats', and 'System' tabs. The 'Threats' tab is active, showing a 'Detections' summary with 17 total findings and a 'Cases' section with a 'Something went wrong' message. Below this, a table titled 'Top 50 high-priority detections (Last 72 hours) 17 items' lists individual detections with columns for Risk score, Severity, Date, Name, User, Source, and Destination. The table shows several detections with a Risk score of 54 and High severity, all originating from the asset DRB-PF312QGZ.

Risk score	Severity	Date	Name	User	Source	Source IP	Destination
54	High	19:14:36	OnWrite-Pre...	DavidBotte	DRB-PF312Q...	10.2.0.2	—
54	High	19:14:32	OnWrite-Pre...	DavidBotte	DRB-PF312Q...	10.2.0.2	—
54	High	19:14:32	OnWrite-Pre...	DavidBotte	DRB-PF312Q...	10.2.0.2	—
54	High	19:14:32	OnWrite-Pre...	DavidBotte	DRB-PF312Q...	10.2.0.2	—
54	High	19:14:33	OnWrite-ML.S...	DavidBotte	DRB-PF312Q...	10.2.0.2	—
54	High	19:14:33	OnWrite-ML.S...	DavidBotte	DRB-PF312Q...	10.2.0.2	—
54	High	19:14:32	OnWrite-ML.S...	DavidBotte	DRB-PF312Q...	10.2.0.2	—
50	High	19:14:32	MLSensorFP...	DavidBotte	DRB-PF312Q...	10.2.0.2	—

Closing the Loop Took Two More Fixes

- Extended the script to update and close alerts via API — not just read them

Wrong parameter name (ids vs. composite_ids)

```
PS C:\Users\DavidBotte\OneDrive - MSFT\Falcon> & C:\Users\DavidBotte\AppData\Local\Microsoft\WindowsApps\python3.13.exe "c:/Users/DavidBotte/OneDrive - MSFT/Falcon/falcon_triage_alerts.py"
Found 10 alert(s) to triage.

Failed to update alerts: [{'code': 400, 'message': 'at least one identifier should be present in the request'}]
PS C:\Users\DavidBotte\OneDrive - MSFT\Falcon> █
```

Wrong status value — fixed using the exact values from the API's own error message

```
PS C:\Users\DavidBotte\OneDrive - MSFT\Falcon> & C:\Users\DavidBotte\AppData\Local\Microsoft\WindowsApps\python3.13.exe "c:/Users/DavidBotte/OneDrive - MSFT/Falcon/falcon_triage_alerts.py"
Found 10 alert(s) to triage.

Failed to update alerts: [{'code': 400, 'message': 'invalid status for update_status action param. must be one of ["closed:A closed alert", "new:A new alert", "in_progress:An in progress alert", "reopened:A reopened alert"]}']
PS C:\Users\DavidBotte\OneDrive - MSFT\Falcon> █
```

Automated Triage, Confirmed

- All 10 alerts closed via API with a status update and an audit comment explaining the authorized test — the same closing discipline used in the Sentinel lab, done here through code

```
PS C:\Users\DavidBotte\OneDrive - MSFT\Falcon> & C:\Users\DavidBotte\AppData\Local\Microsoft\WindowsApps\python3.13.exe "c:/Users/DavidBotte/OneDrive - MSFT/Falcon/falcon_triage_alerts.py"
Found 10 alert(s) to triage.

Successfully triaged 10 alert(s):
- 7391a4d69e3d4bb2aa3ccb993fedd55f:ind:137e3b40aa59408e97be921d67fe2fda:10072546568-10417-642320
- 7391a4d69e3d4bb2aa3ccb993fedd55f:ind:137e3b40aa59408e97be921d67fe2fda:10072546568-10417-626448
- 7391a4d69e3d4bb2aa3ccb993fedd55f:ind:137e3b40aa59408e97be921d67fe2fda:8884985515-10127-613904
- 7391a4d69e3d4bb2aa3ccb993fedd55f:ind:137e3b40aa59408e97be921d67fe2fda:10034639130-10127-613392
- 7391a4d69e3d4bb2aa3ccb993fedd55f:ind:137e3b40aa59408e97be921d67fe2fda:10034639130-5734-593936
- 7391a4d69e3d4bb2aa3ccb993fedd55f:ind:137e3b40aa59408e97be921d67fe2fda:10034639130-4748-605200
- 7391a4d69e3d4bb2aa3ccb993fedd55f:ind:137e3b40aa59408e97be921d67fe2fda:10034639130-5738-609296
- 7391a4d69e3d4bb2aa3ccb993fedd55f:ind:137e3b40aa59408e97be921d67fe2fda:10034639130-5733-532752
- 7391a4d69e3d4bb2aa3ccb993fedd55f:ind:137e3b40aa59408e97be921d67fe2fda:10034639130-5733-546064
- 7391a4d69e3d4bb2aa3ccb993fedd55f:ind:137e3b40aa59408e97be921d67fe2fda:10034639130-4748-482576
PS C:\Users\DavidBotte\OneDrive - MSFT\Falcon> █
```

API client updated with write access, scoped only to what triage required

Testing Across Multiple Tactics

- Ran two more Atomic Red Team techniques, covering different MITRE tactics than the original test
- T1082 (Discovery): ran cleanly, no alert — expected, since a native systeminfo command alone isn't distinct enough to flag
- T1112 (Defense Evasion): blocked by Falcon's prevention layer AND still generated a High severity alert — prevention and detection working together, not just one or the other

```
PS C:\Users\DavidBotte\OneDrive - MSFT\Falcon> python falcon_list_alerts.py

Found 10 alert(s):

-----
Hostname                Severity  Status    Created                                Tactic
-----
DRB-PF312QGZ            High      new       2026-08-01T01:15:54.304645005Z        Defense Evasion
DRB-PF312QGZ            Informational new       2026-08-01T01:15:54.3011847Z          Execution
DRB-PF312QGZ            Informational new       2026-08-01T01:12:54.635607071Z        Execution
DRB-PF312QGZ            Medium    new       2026-08-01T00:51:06.189747275Z          Execution
DRB-PF312QGZ            Medium    new       2026-08-01T00:51:06.154824638Z          Execution
DRB-PF312QGZ            Medium    new       2026-08-01T00:51:05.220115589Z          Machine Learning
DRB-PF312QGZ            High      new       2026-08-01T00:51:05.201007623Z          Machine Learning
DRB-PF312QGZ            Low       new       2026-08-01T00:51:05.187563568Z          Malware
DRB-PF312QGZ            Low       new       2026-08-01T00:51:05.186290972Z          Malware
DRB-PF312QGZ            High      new       2026-08-01T00:51:05.184135574Z          Machine Learning
PS C:\Users\DavidBotte\OneDrive - MSFT\Falcon>
```

WHAT THIS DEMONSTRATES

Skills applied in this lab

- **EDR & API Integration**

CrowdStrike Falcon, FalconPy SDK, OAuth2 authentication

- **Real API Debugging**

Diagnosed and fixed 4 distinct real errors, not scripted ones

- **Least Privilege**

Scoped API client access to exactly what each task needed

- **Adversary Simulation**

MITRE ATT&CK technique execution via Atomic Red Team

- **Automated Triage & Response**

Closed the loop: read, triaged, and contained hosts via API — with safety-first dry-run design

- **Verification Discipline**

Cross-checked script output against the console directly